

Spotify Aura: GenAI-Based Personalized Music Discovery Prototype

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This report presents Spotify Aura, an unofficial Spotify-inspired GenAI product prototype for personalized music discovery. The system converts user mood, intent, time, language preference, social context and consent signals into an adaptive listening session. It is designed as a deployable React + Vite frontend prototype for Vercel, with a believable mobile music interface, explainable recommendations, group Jam logic and privacy controls.

Project Field	Details
Project Type	Frontend product prototype
Core Idea	Turn natural-language listening intent into a complete AI-generated session
Technology	React, Vite, CSS, Lucide React, Vercel
External API Use	None in prototype version
GitHub Repository	https://github.com/satyam9140/Spotify_AI.git

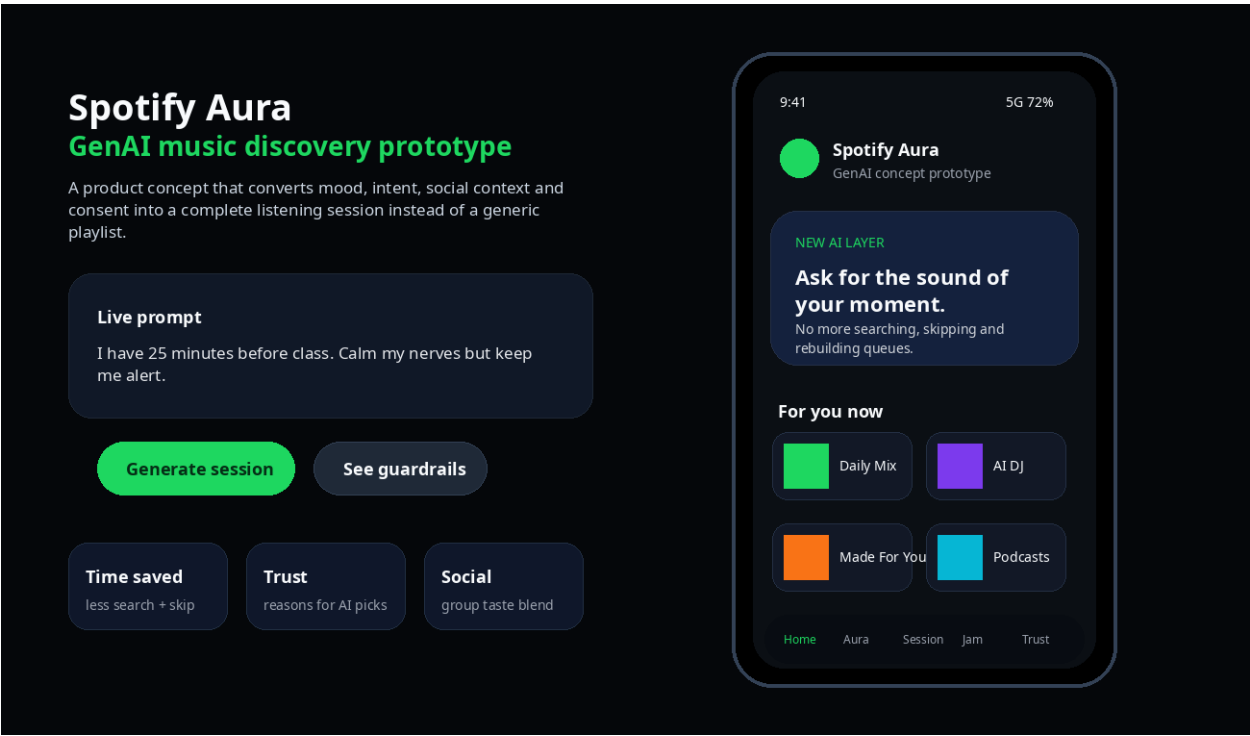


Figure 1: Spotify Aura prototype overview showing the prompt panel, session generation controls and Spotify-style mobile interface.

Introduction

Music streaming platforms already provide strong recommendation systems, but the user experience is still often passive. Users may need to search manually, skip repeatedly, switch between playlists or describe their exact situation through trial and error. Spotify Aura reframes music discovery around intent: instead of asking the user to choose a genre or playlist, it lets the user describe the current moment in natural language.

The prototype demonstrates how a GenAI layer can sit above a recommender system. It interprets the user prompt, creates a session with a defined tone, generates a track queue, explains why each track appears and gives the user clear control over memory and trust settings.

Objectives

- Build a prompt-driven listening experience where users can ask for the sound of their current moment.
- Generate a believable music session with title, tone, duration, tracks and selection reasons.
- Demonstrate group listening through a Jam Agent that balances multiple users and constraints.
- Include a privacy and trust layer with opt-in memory and transparent AI decisions.
- Keep the project lightweight, reproducible and ready for Vercel deployment without API keys.

Problem Statement Relevance

The central product problem is not just recommending songs. The stronger problem is reducing the effort required to convert a personal context into the right listening experience. Spotify Aura answers this with a user-facing GenAI interaction pattern: prompt, interpret, generate, explain, control and share.

Methodology and System Architecture

The workflow begins when the user enters a natural-language prompt. The prototype then maps the prompt to a session category such as focus, rainy-night Hindi indie, gym/training or group road-trip mode. The selected mode determines tone, queue structure, discovery balance and explanation messages. The interface then presents the generated session in a mobile-first music-player layout.

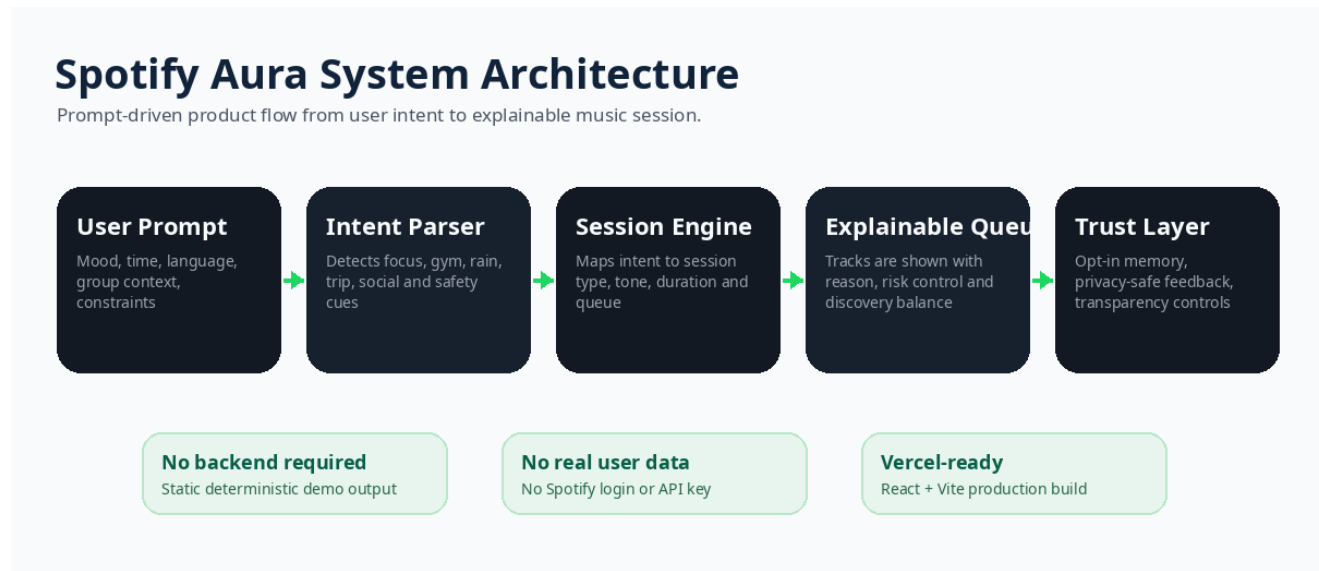


Figure 2: System architecture from prompt capture to trust-controlled explainable session output.

Method Flow

Stage	Description
Input Capture	User describes mood, time, activity, language preference or group context.
Intent Detection	Prompt keywords are mapped into focus, rainy, gym or social session modes.
Session Generation	The prototype creates a session title, tone, reasoning block, metrics and queue.
Explanation Layer	Each recommended track includes a short reason for selection.
Control Layer	User can access memory consent, trust controls and group-sharing actions.

Core Features and Product Flow

Spotify Aura is intentionally scoped as a strong product prototype rather than a fake full-stack clone. The most important feature is the complete end-to-end demo flow: a judge can open the deployed link, enter a prompt, receive a generated session, review explanations, inspect group Jam behavior and understand the privacy controls.

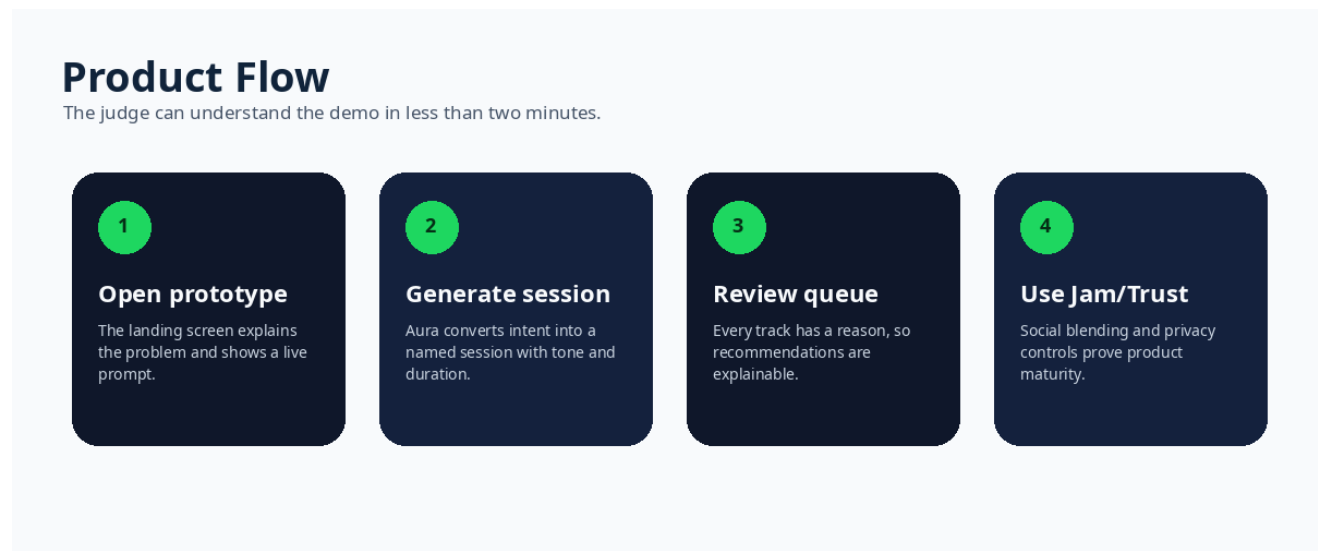


Figure 3: Recommended demo flow for presenting the product in a judging or submission environment.

Feature Summary

Feature	Implementation Value
Prompt-to-Session AI Flow	Shows GenAI relevance through natural-language music discovery.
Spotify-like Mobile UI	Makes the concept instantly understandable and product-realistic.
Generated Queue with Reasons	Improves trust by explaining why each song was selected.
Group Jam Agent	Handles real social listening problems such as taste conflict and content filtering.
Trust and Memory Controls	Shows responsible personalization instead of hidden data use.

AI Session Generation Output

The session-generation screen is the strongest part of the prototype. It converts a user situation into an actionable music session. A sample prompt such as "I have 25 minutes before class. Calm my nerves but keep me alert" produces a focus-reset session with a calm but alert tone, controlled discovery, low-distraction queue design and session metrics.

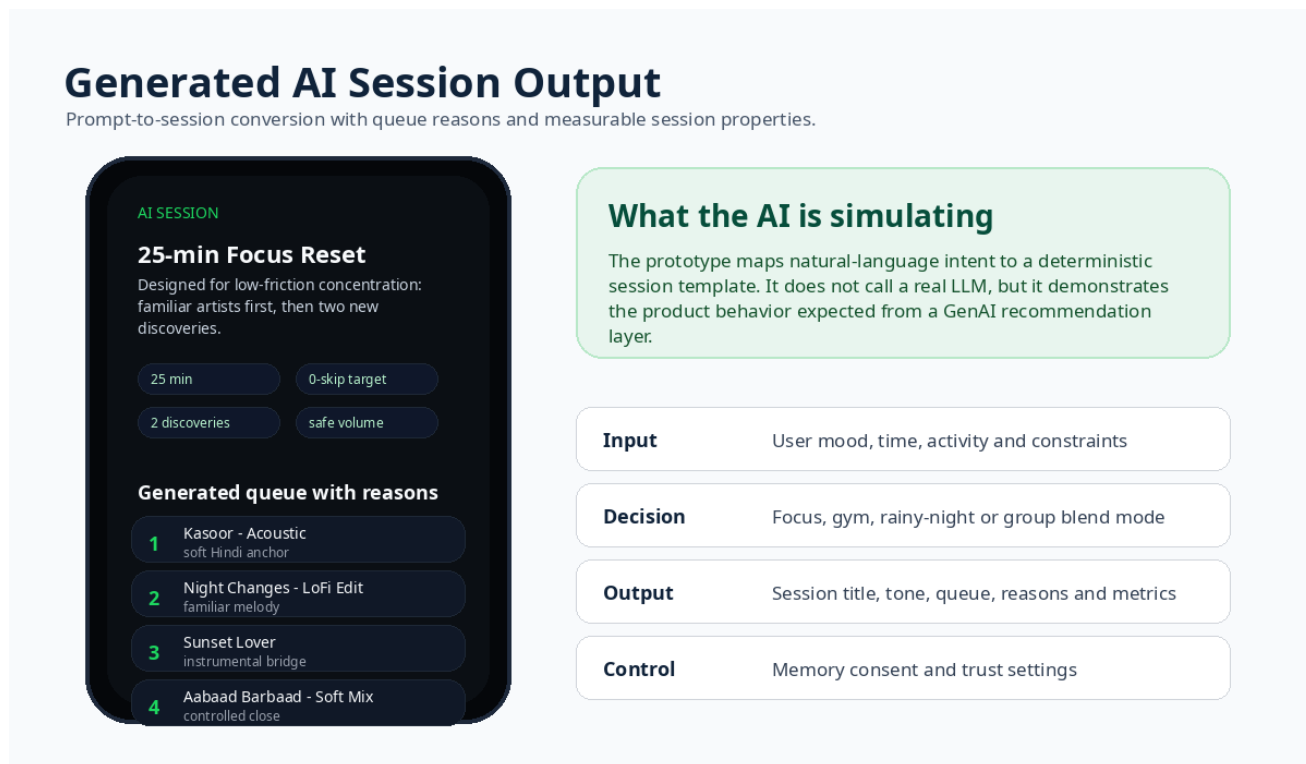


Figure 4: AI-generated focus session showing queue reasoning, session metrics and explainable recommendation behavior.

Generated Session Components

Component	Example in Prototype
Session Title	25-min Focus Reset
Tone	Calm, alert, familiar first
Reasoning	Starts with familiar anchors and adds controlled discovery.
Metrics	Duration, skip target, discovery count and safe-volume cue.
Track Explanation	Each track includes a short reason for why it fits the session.

Group Jam Agent and Trust Controls

The Group Jam Agent improves the product concept because music listening is often social. Instead of allowing one person to dominate the shared queue, the agent simulates negotiation between user preferences. In the sample flow, Hindi, Afrobeats and content restrictions are balanced into one road-trip session.

The Trust Layer is equally important. Personalization can become uncomfortable if users do not understand what the AI remembers or why a recommendation appears. Spotify Aura therefore presents memory as opt-in, keeps explanation visible and labels AI-generated behavior clearly.

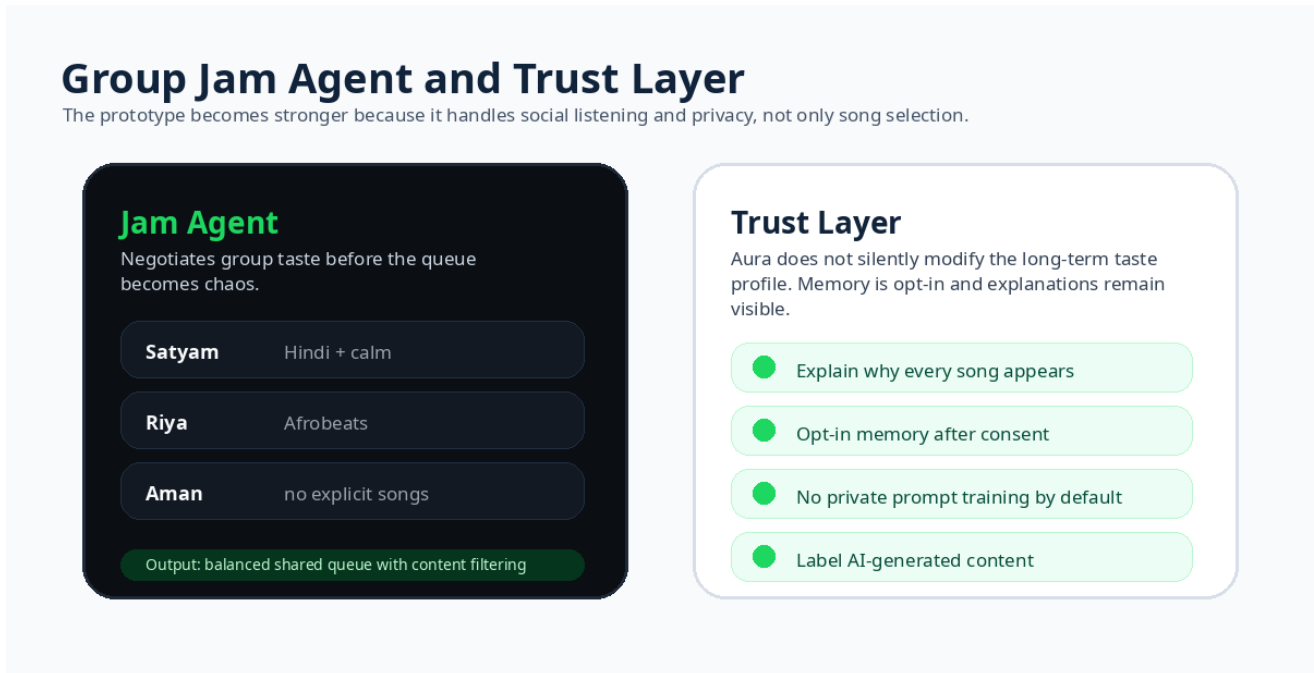


Figure 5: Group Jam and Trust Layer concept screens showing social taste blending and privacy-safe personalization controls.

Trust Design Principles

- Memory should be off by default unless the user explicitly gives consent.
- Every AI recommendation should be explainable in simple language.
- Synthetic or AI-generated audio/content should be disclosed.
- Sensitive prompts should be handled cautiously and not treated as normal music-preference data.

Technical Implementation and Reproducibility

The repository follows a standard frontend prototype structure. The app is built with React, served through Vite and prepared for deployment through Vercel. The prototype does not require backend services, database setup, OAuth configuration or external music API keys, which makes it reliable for demonstration and academic submission.

Technology Stack

Layer	Technology / Decision
Frontend Framework	React
Build Tool	Vite
Styling	CSS with dark mobile-first product interface
Icons	Lucide React
Deployment Target	Vercel
Backend	Not required for prototype
External APIs	Not used in current demo

Run Commands

Purpose	Command
Install dependencies	npm install
Start development server	npm run dev
Create production build	npm run build
Preview production build	npm run preview
Deploy on Vercel	Import GitHub repo, build command: npm run build, output: dist

Repository Link

https://github.com/satyam9140/Spotify_AI.git

Evaluation Summary, Limitations and Future Scope

Because Spotify Aura is a product prototype, the correct evaluation is not a fake accuracy score. The project should be judged on clarity of product problem, GenAI relevance, usability, implementation completeness, deployment readiness and responsible AI thinking.

Prototype Capability Summary

Evaluation Area	Status
Clear Product Problem	Implemented - solves intent-to-session discovery friction.
Prompt-Based GenAI Flow	Implemented - natural-language prompt generates a music session.
Explainable Recommendations	Implemented - track-level reasons are shown.
Social Listening Feature	Implemented - Group Jam Agent balances preferences.
Trust and Privacy Thinking	Implemented - memory consent and transparency controls.
Deployment Readiness	Implemented - Vercel-ready React + Vite frontend.
Production Backend	Not implemented - acceptable for prototype scope.

Current Limitations

- The current prototype does not use the real Spotify Web API.
- The AI output is simulated and deterministic rather than generated by a live LLM.
- There is no authentication, user database or persistent memory.
- No real user listening metrics are collected in the frontend-only demo.

Future Scope

- Integrate Spotify Web API for real playlist, track and profile data.
- Connect a live LLM for real prompt interpretation and playlist generation.
- Add user accounts, session history and privacy-safe preference memory.
- Add collaborative queue voting, skip analytics and group satisfaction scoring.
- Run usability tests to measure search-time reduction, skip-rate reduction and user trust.

Conclusion

Spotify Aura successfully demonstrates how GenAI can become an interaction layer on top of a music recommendation platform. The prototype is strong because it does not simply copy a streaming interface; it shows a complete product idea: intent capture, session creation, explainable recommendations, group listening and trust controls. With real API integration and live model output, this concept can be extended into a production-grade personalized music discovery system.

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